

Properties of Fourier Series

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1 Properties of Fourier Series

A function $f(x)$ can be written as the series shown in ??, known as a Fourier Series. The following provides derivation of the properties of Fourier Series.

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx) \quad (1)$$

1.0.1 Side Note: Trigonometric Identities of addition

Recall equations ????????

$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b) \quad (2)$$

$$\cos(a-b) = \cos(a)\cos(b) + \sin(a)\sin(b) \quad (3)$$

$$\sin(a+b) = \sin(a)\cos(b) + \sin(b)\cos(a) \quad (4)$$

$$\sin(a-b) = \sin(a)\cos(b) - \sin(b)\cos(a) \quad (5)$$

From equations ???????? we obtain ??????

$$\cos(a)\cos(b) = \frac{1}{2}(\cos(a+b) + \cos(a-b)) \quad (6)$$

$$\sin(a)\cos(b) = \frac{1}{2}(\sin(a+b) + \sin(a-b)) \quad (7)$$

$$\sin(a)\sin(b) = \frac{1}{2}(\cos(a-b) - \cos(a+b)) \quad (8)$$

1.0.2 Integrals

Since $\sin(nx)$ is an odd function, its integral over a symmetric region is zero.

$$\int_{-\pi}^{\pi} \sin(nx) dx = 0 \quad (9)$$

Since n is an integer, and assuming $n > 0$, we obtain the following.

$$\int_{-\pi}^{\pi} \cos(nx) dx = \frac{2\sin(\pi n)}{n} = 0 \quad (10)$$

Since $\sin(mx)\cos(nx)$ is an odd function, its integral over a symmetric region is zero.

$$\int_{-\pi}^{\pi} \sin(mx)\cos(nx)dx = 0 \quad (11)$$

Use ?? to write the following.

$$\begin{aligned} \int_{-\pi}^{\pi} \cos(mx)\cos(nx)dx &= \frac{1}{2} \int_{-\pi}^{\pi} \cos((n+m)x)dx + \frac{1}{2} \int_{-\pi}^{\pi} \cos((n-m)x)dx \\ &= \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases} \quad (12) \end{aligned}$$

Using ?? we obtain the following.

$$\begin{aligned} \int_{-\pi}^{\pi} \sin(mx)\sin(nx)dx &= \frac{1}{2} \int_{-\pi}^{\pi} \cos((n-m)x)dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos((n+m)x)dx \\ &= \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases} \quad (13) \end{aligned}$$

1.0.3 a_0 coefficient

Integrate both sides of ?? to obtain:

$$\int_{-\pi}^{\pi} f(x)dx = \int_{-\pi}^{\pi} a_0 dx + \sum_{n=1}^{\infty} (a_n \int_{-\pi}^{\pi} \cos(nx)dx + b_n \int_{-\pi}^{\pi} \sin(nx)dx)$$

Using ???? the sum terms become zero.

$$= \int_{-\pi}^{\pi} a_0 + 0 + 0 = a_0 x|_{-\pi}^{\pi} = a_0 \pi + a_0 \pi = 2\pi a_0$$

And thus we obtain

$$a_0 = \frac{\int_{-\pi}^{\pi} f(x)dx}{2\pi} \quad (14)$$

1.0.4 a_n coefficients

Multiply both sides of ?? by $\cos(mx)$ to obtain:

$$f(x)\cos(mx) = a_0\cos(mx) + \sum_{n=1}^{\infty} a_n\cos(nx)\cos(mx) + b_n\sin(nx)\cos(mx)$$

Integrate both sides.

$$\int_{-\pi}^{\pi} f(x)\cos(mx)dx = a_0 \int_{-\pi}^{\pi} \cos(mx)dx + \sum_{n=1}^{\infty} (a_n \int_{-\pi}^{\pi} \cos(nx)\cos(mx)dx + b_n \int_{-\pi}^{\pi} \sin(nx)\cos(mx)dx)$$

$$\rightarrow \int_{-\pi}^{\pi} \cos(mx) dx = 0 \text{ (from ??)}$$

$$\rightarrow \sum_{n=1}^{\infty} (a_n \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx = \int_{-\pi}^{\pi} a_m \cos(mx) \cos(mx) + \sum_{n=1, n \neq m}^{\infty} \int_{-\pi}^{\pi} \cos(nx) \cos(mx),$$

$$\text{which (from ??) is equal to } a_m \pi + \sum_{n=1, n \neq m}^{\infty} 0 = a_m \pi.$$

$$\rightarrow \int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx = 0 \text{ (from ??). Thus we obtain:}$$

$$\int_{-\pi}^{\pi} f(x) \cos(mx) dx = 0 + a_m \pi + 0$$

And thus:

$$a_m = \frac{\int_{-\pi}^{\pi} f(x) \cos(mx) dx}{\pi}$$

Which, by changing the index from m to n gives:

$$a_n = \frac{\int_{-\pi}^{\pi} f(x) \cos(nx) dx}{\pi} \quad (15)$$

1.0.5 b_n coefficients

Using similar arguments to ?? we obtain:

$$b_n = \frac{\int_{-\pi}^{\pi} f(x) \sin(nx) dx}{\pi} \quad (16)$$